```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
# Create sample salary data
df=pd.DataFrame({
    "YearsExperience": [1.1, 1.5, 2, 2.2, 2.9, 3.2, 3.7, 4,
                        4.5, 4.9, 5.3, 5.9, 6.2, 6.8, 7.1,
                        7.9, 8.2, 8.7, 9, 9.5, 10, 10.5],
    "Salary": [39343, 37731, 43525, 39891, 56642, 54445, 57189,
               55794, 61111, 67938, 66029, 81363, 93940, 91738,
               98273, 101302, 113812, 109431, 105582, 116969,
               122391, 121872]
})
print(df.head())
print(df.info())
print(df.describe())
features = df[["YearsExperience"]].values
label = df[["Salary"]].values
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(x_train, y_train)
print("Training Score:", model.score(x_train, y_train))
print("Testing Score:", model.score(x_test, y_test))
print("Coefficient:", model.coef_)
print("Intercept:", model.intercept_)
plt.scatter(features, label)
plt.plot(features, model.predict(features), color="red")
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Linear Regression - Salary Prediction")
plt.show()
years = np.array([[5.0]])
salary = model.predict(years)
print("Estimated Salary for 5 years of experience:", salary[0][0])
```

```
   YearsExperience  Salary
0              1.1   39343
1              1.5   37731
2              2.0   43525
3              2.2   39891
4              2.9   56642
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 22 entries, 0 to 21
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
```

```
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 0   YearsExperience  22 non-null     float64
 1   Salary           22 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 484.0 bytes
None
       YearsExperience         Salary
count        22.000000      22.000000
mean          5.686364   78923.227273
std           2.914381   29324.289461
min           1.100000   37731.000000
25%           3.325000   56006.000000
50%           5.600000   74650.500000
75%           8.125000  104512.000000
max          10.500000  122391.000000
Training Score: 0.965147561174963
Testing Score: 0.9761526830926572
Coefficient: [[10052.26447106]]
Intercept: [21569.82824351]
```




```
Estimated Salary for 5 years of experience: 71831.1505988024
```